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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Bailey, Michael (University of Washington) Carter, Stephen (University of Washington) Vaezy, Shahram (University of Washington) Khohklova, Vera (M.V.Lomonosov Moscow State University) Sapozhnikov, Oleg (M.V.Lomonosov Moscow State University)		
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Task Description:

(1) Original Aims
The principal, long-term objective of this proposed effort is to develop a smart medical device that would be lightweight, portable, FDA-approved, under commercial development, and capable of addressing various aspects of four risks described in the Bioastronautics Roadmap [NASA/SP-2005-6113]. In particular, we seek to address under Risk 18, the risk of blunt internal trauma and internal bleeding resulting, for example, from a collision in microgravity between an astronaut and a heavy object. The device we propose would be capable of detecting internal bleeding and inducing transcutaneous hemostasis. Under Risks 18 and 28, as recent experiences in Antarctica demonstrate [Nielsen, 2001], malignant tumors that require some form of surgery may well appear without warning, especially in the high-radiation load of outside LEO, even when extensive pre-screening is undertaken. The device we propose would be capable of performing bloodless ablation of detectable tumors without fear of metastasis. Under Risk 4, the formation of renal calculi has been suggested by Schneider, et al. [1994] and by Ball and Evans [2001] as a likely result in extended space flight; indeed, a precedent exists of early mission termination due to a ureteral stone [Pietrzyk, et al, 2003]. The device we propose would be capable of detecting and sufficiently comminuting a renal calculus for it to pass without obstruction through the urinary system.

Specific Aim 1. To expand the capability of our existing smart medical device for transcutaneous acoustic hemostasis to include a fully integrated ultrasound detection, targeting, and therapy system.

Specific Aim 2. To develop the capability of the integrated image-guided therapy system in SA 1 to target and ablate specific tissue volumes such as benign and malignant tumors.

Specific Aim 3. To develop the capability of the integrated image-guided therapy system in SA 1 to target and comminute renal calculi.

(2) Key Findings

In this particular effort and through related work, we have developed a variety of high intensity focused ultrasound (HIFU) systems to investigate acoustic hemostasis, tissue ablation, and kidney stone comminution. A broad array of our successful specific applications of these systems has been published. These devices continue to be refined to make them easier to use and acceptable for the FDA approval.

In the first three years of the grant, our focus has been in enhancing the efficacy of our laboratory prototypes toward accurate treatment planning and clinical approval. The NSBRI funded research has led to the following main findings that are elaborated in the claims section and in the publications listed in this report:

• Two frequencies can accelerate heating and hemostasis.

• Addition of microbubble contrast agents can accelerate heating and hemostasis.

• Nonlinear acoustic propagation can be exploited to accelerate heating.

• Boiling caused by nonlinear acoustic waves can be used for imaging, targeting, and treatment planning.

• A model was developed to account for heating by cavitation bubbles.

• A model was developed to optimize heating enhancement through nonlinear acoustics.

• A high-power system was used to complete full thickness acoustic hemostasis of bleeding in organs, and to occlude vessels as large as 1 cm in diameter.

• A new method, as well as the associated instrumentation, was created to accurately calibrate HIFU systems.

In addition, progress was made in the study of HIFU to comminute kidney stones, in efforts to guide treatment, and toward animal and human clinical trials. The main findings are listed here. • High power, two frequency systems have proven effective in eroding kidney stones.

• A model was developed and tested to predict kidney stone fracture.

• Tools and a technique were developed to detect kidney stones and determine comminution.

• A patent was obtained to synchronize imaging and HIFU.

• An IDE (Investigational Device Exemption) has been submitted to the FDA for use of a clinical HIFU system to treat pancreatic cancer in patients. [The NSBRI work was mostly to support acoustic characterization tests of the device.]

(3) Impact on the hypotheses, requirements, objectives, and specific aims of the original proposal

Significant advancement toward higher TRL/CRL levels has been achieved as our technologies and methodologies improve. In particular, we are leveraging our work with other funding agencies to significantly improve our technology and can anticipate human clinical trials in the future. Our principal effort in year three was to improve our capabilities in diagnosis, targeting, and monitoring of treatment.

(4) Proposed Research Plan

Half a dozen manuscripts are currently in the composition or review process and it is a priority to see them all published. Our efforts for year four will focus on the following topics: Our unique understanding of the acoustic parameters (such as acoustic intensity) delivered in vivo will be used to establish a therapeutic dose for each of the bioeffects we study. We will conduct a variety of tests of an integrated automated imaging and therapy acoustic hemostasis system. This system, supported by DARPA and in partnership with Philips, uses conventional Doppler imaging to establish a unique acoustic signature for a bleeding vessel—as distinct from a normal one—which would then permit autonomous targeting and therapy. We will also leverage our NIH-supported work on tissue ablation to advance our knowledge of the requirements for HIFU therapy; indeed, we expect to be in human clinical trials before the end of the year with a commercial HIFU system that is present in our laboratory. Lastly we will expand our methods to detect kidney stones using Doppler ultrasound and our patented system to determine the size of the comminuted stone fragment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	We have proposed to develop a smart medical device that would have specific applications to the risks associated with space travel, and due to the major constraints and uniqueness of this environment, this device, as presently conceived, would most probably not, in itself, be of value to Earth-based clinical medicine. However, the various technologies that are to be incorporated into this device have many important clinical applications. In particular, we seek to build a lightweight, image-guided therapy device that would have the capability to ablate tumors, stop bleeding, and to detect, as well as to comminute, small kidney stones—all in a single device. These medical conditions are commonplace also on Earth, and the technological spin-offs of this technology are well developed. Uncontrolled, occult bleeding due to trauma is a major Earth-based risk. In a multi-center review of 72,151 trauma admissions at a civilian health center, 62% of those requiring emergent operative treatment were in shock on arrival in the operating room, and the actual cause of death was credited to bleeding in 82% of the cases [Hoyt, et al., 1994]. Indeed, exsanguination is often reported as the major cause of mortality (~50%) in combat casualties [Zajtchuk and Sullivan, 1995]. Accordingly, we have been undertaking a major effort, funded by a variety of federal agencies (but principally by the DoD) and in partnership with industry (e.g., Philips Medical) to develop a portable device that could be used in forward echelons of the battlefield to address the risk of death by exsanguination. We have leveraged this extensive DoD-funded effort in our research for the NSBRI. Furthermore, several companies, in addition to Philips have product development programs underway on this concept, viz., THERUS and ACOUSTX, both based in Seattle, and both of which have licensed technology from the University of Washington (UW). This image-guided therapy approach utilizes the capability of HIFU to rapidly elevate the temperature to coagulative necrosis levels (> 55 oC). The capability to induce coagulative necrosis, particularly under image-guidance, provides a unique opportunity to treat benign and malignant tumors. Two Seattle-based start-up companies, MIRABILIS and UST, have licensed UW technology to develop products that would use image-guided HIFU to ablate tumors. Our NSBRI Research Team has also assisted YUANDE, a Chinese company, in its application to the FDA for an IDE (Investigative Device Exemption) to treat patients with advanced stage pancreatic cancer. Our laboratory has had considerable experience in the treatment of stone disease; we are now participating in the 14th year of an NIH Program Project Grant to study the physical basis of stone comminution and tissue damage in shock-wave lithotripsy. Between 5% and 15% of the general population will develop a urinary calculus during their lifetime [Consensus conference. JAMA 1988;260:977] and will account for about 7 to 10 of every 1000-hospital admissions in the United States [Levy et al. 1995]. One of our major contributions to lithotripsy research has been the determination that acoustic cavitation plays a major role in stone comminution [McAteer, et al., 2003; Pishchalnikov, et al., 2002; Sapozhnikov, et al., 2002; Evan, et al., 2002]. Our NSBRI project builds upon this extensive experience to develop a system for detecting and comminuting a kidney stone in the unique and remote environment of space. Because fluoroscopy will (probably) not be available in extended space travel missions, our ultrasound-based detection and comminution approach has been developed. In performing this research, we have developed unique techniques and methodologies for monitoring the progress of lithotripsy treatment that appear to have immediate Earth-based applications.
Task Progress:	Tasks 1a-d: To develop a probe applicator containing an embedded image probe together with an annular array capable of delivering HIFU at therapy levels...To integrate the probe applicator with a human interface and power unit... To perform a series of experiments in ex vivo ... to induce acoustic hemostasis transcutaneously. An integrated applicator has been developed and tested. The device is described in the manuscript by Owen and in the patent application. Our NSBRI Fellow has published on progress working with our array system. 2a-c. To develop a real-time guidance control and targeting system that permits HIFU lesions to be produced at specific sites ... To develop a lesion-imaging technique to permit real-time monitoring of the HIFU lesion produced ... of interest in ex vivo and phantom tissues. These 3 tasks have essentially been completed with our simple image-guided HIFU therapy system. The system relies on the appearance of an echogenic region on B-mode or Doppler imaging to indicate the location of the HIFU focus. This image-guidance technique is used the clinical system we are testing and our patent application describes a method and a device to permit imaging while using HIFU. Our recent paper in the J. Acoustical Soc. America describes a targeting approach that uses a nonlinear acoustics approach to rapidly produce boiling, which in turn produces an echogenic spot at the HIFU focus. 2d: To develop a therapy protocol based upon a validated numerical model... A model based on linear acoustic propagation was developed in FEMlab for use in temperature monitoring described above. A more refined model incorporating nonlinear acoustic propagation and using a finite difference method is being tested in in vitro and in vivo experiments. A number of publications, describing these techniques, have been published or are in press. 3a-c. To determine through both modeling and in vitro experimentation the acoustic protocols required to produce cavitation levels sufficient to comminute artificial kidney stones... To measure the acoustic pressure amplitudes and intensities that are produced... To determine the minimal sizes of renal calculi that can be detected... Dr. Bailey spent approximately 1 month in Japan working with our collaborators on using their two-frequency system to comminute stones; a Postdoc from our Japanese collaborators' lab joined us for 4 months. In 2006 we reported optimized stone comminution within a calibrated exposure range. 3d. To demonstrate the capability of the integrated system...A fully integrated system for stone comminution has not yet been completed; however, many of the essential sub-elements have been produced and tested. Our former Postdoc is developing a similar system at Hitachi. We have discovered a “twinkling artifact” method to detect stones with diagnostic ultrasound and have filed a patent application for a method to monitor stone comminution.
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